

2	(a)	Since $g \propto M/r$, $\frac{g_M}{g_E} = \left(\frac{M_M}{M_E}\right) \left(\frac{r_E}{r_M}\right)^2$ $= (1/81)(3.7)^2$ $= 0.169$ Therefore, g due to Moon = 0.169×9.81 $= 1.66 \text{ N kg}^{-1}$		M1 C1 A1
	(b)	$\frac{GM_E}{r^2} = \frac{GM_M}{(3.84 \times 10^8 - r)^2}$ $81 = \frac{r^2}{(3.84 \times 10^8 - r)^2}$ $r = 3.46 \times 10^8 \text{ m}$		M1 A1
	(c)	(i)	$\omega = 2\pi/T$ $= 2\pi/(27.3 \times 24 \times 60 \times 60)$ $= 2.66 \times 10^{-6} \text{ rad s}^{-1} [1]$	A1
		(ii)	Gravitational acceleration provides centripetal acceleration. $GM/r^2 = r\omega^2$ $M = r^3\omega^2/G$ $= (3.84 \times 10^8)^3 \times (2.66 \times 10^{-6})^2 / 6.67 \times 10^{-11}$ $= 6.0 \times 10^{24} \text{ kg}$	M1 A1
		(iii)	Mass of Moon = 1/81 Mass of Earth $F = GM_E M_M / r^2$ $= [6.67 \times 10^{-11} (6.0 \times 10^{24})^2 / 81] / (3.84 \times 10^8)^2 [1]$ $= 2.01 \times 10^{20} \text{ N}$	M1 A1
		(iv)	The acceleration the moon causes on the Earth is very small as the Earth has a large mass. $a = F/m = 2.01 \times 10^{20} / 6.0 \times 10^{24} = 3.35 \times 10^{-5} \text{ m s}^{-2}$	B1 A1